

"I Can Only Guess Where She's Looking": Challenges and Design Strategies for Supporting Outdoor Informal Learning Between Parents with Visual Impairments and Their Sighted Children

Yutong Jiang*
College of Design and Innovation
Tongji University
Shanghai, China
jiangyytong@outlook.com

Qingyun Zheng*
College of Design and Innovation
Tongji University
Shanghai, China
zqy2001425@163.com

Zixuan Zhang*
College of Design and Innovation
Tongji University
Shanghai, China
zxx1021@tongji.edu.cn

Qian Guo*
College of Design and Innovation
Tongji University
Shanghai, China
m18516316016@163.com

Jiaying Xu*
College of Design and Innovation
Tongji University
Shanghai, China
xujiaying@tongji.edu.cn

Shuchang Xu
Department of Computer Science and
Engineering
The Hong Kong University of Science
and Technology
Hong Kong, China
sxuby@connect.ust.hk

Guanhong Liu[†]
College of Design and Innovation
Tongji University
Shanghai, China
892489960@qq.com

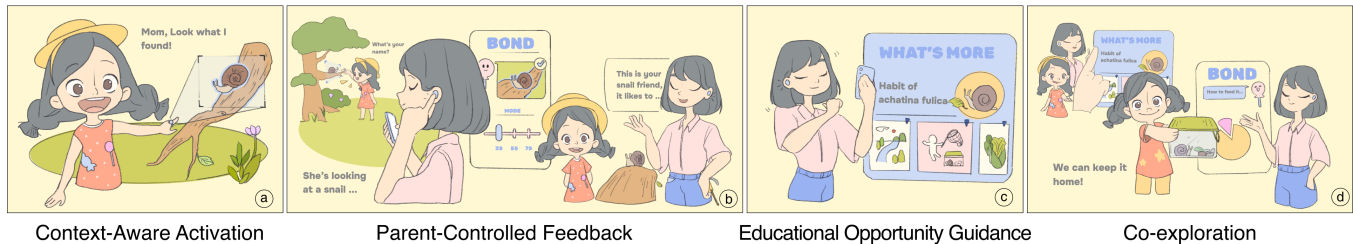


Figure 1: Four key scenarios demonstrate how our system supports PVI and their sighted children during outdoor informal learning.

Abstract

Parents with visual impairments (PVI) face unique challenges when engaging in outdoor informal learning with their sighted children due to visual asymmetries. Based on the study of 11 families, we identified three recurring parenting patterns—over-involved, hands-off, and limited support—shaped by cognitive, perception and social identity. To facilitate informal outdoor learning, we propose

Symbiotic Education Model to support child-led exploration while enabling PVI to interpret, respond to, and co-construct meaning, offering new directions for inclusive parent-child interaction design in visually asymmetric contexts.

CCS Concepts

• **Human-centered computing** → **Accessibility**.

Keywords

Parents with Visual Impairments; Informal Learning; Parent-child Interaction; Inclusive Design

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*These authors contributed equally to this research.

[†]Corresponding author.

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1 Introduction

According to the World Health Organization, over 2.2 billion people worldwide have some form of visual impairments. China has one of the largest populations of people with visual impairments, with more than 17 million individuals affected as of 2022 [25], many of whom are of parenting age. As most causes of visual impairment are non-hereditary - and with increasing access to early screening and intervention - most families consist of the parents with visual impairments (PVI) and the sighted children. Despite inclusive policies [18] and advances in assistive technologies that support daily functioning, visual asymmetry between parents who rely on tactile and auditory cues and children who navigate primarily through vision continues to impact PVI in engaging in the development and education of their sighted children.

While prior research has examined caregiving [8] and emotional support [23] within the home, little is known about how PVI and their sighted children engage in informal learning. Informal learning refers to the unstructured, everyday acquisition of knowledge, skills, and perceptual awareness through natural and interactive experiences, including early science education and broader developmental growth [14], which is critical to child development. In complex outdoor settings, visual asymmetry can amplify informal learning challenges, making it difficult to establish joint attention [15], weakening interaction quality, and discouraging children's exploratory behaviors.

Existing assistive technologies—such as sensory substitution systems [16, 28, 29], AI description tools [26, 27], and crowdsourced support platforms [2]—are primarily designed for individual perception, offering limited support for the interactional challenges caused by visual asymmetry in PVI-sighted child relationships. While enhancing information access, they often reinforce one-sided assistance or remain too complex for real-time co-engagement. There is a lack of co-perception systems that support PVI in transitioning from passive recipients of mediated input to active participants in outdoor informal learning. To fill this gap, we propose the following research questions:

- RQ1: What barriers do PVI and their sighted children face in outdoor informal learning?
- RQ2: How can design support PVI in engaging more responsively in outdoor informal learning?

We interviewed nine parents and observed two families, identifying three dimensions—cognitive, physical and cultural—which shaped three parenting patterns: over-involved, hands-off, and limited support. In response, we propose a symbiotic education model to help PVI interpret, respond to, and co-explore with their children.

2 Related Work and Background

Parenting inherently involves a series of challenges, including managing safety, emotional bonds, and children's development [8]. For PVI with sighted children, caregiving becomes more complex. First, parent-child communication relies heavily on non-visual cues—such as language, touch, and emotional signals—which often

leads to asymmetrical understanding [5]. Second, role boundaries blur as children sometimes act as “young carers” [24] to assist with navigation or interpretation, which risks placing excessive responsibility on the child [9]. These challenges impact everyday interactions, including shared reading [5, 11, 19, 22], play [23], and appreciating children's artwork [6, 7], as discussed in prior studies.

These challenges intensify when PVI and their children enter visually rich outdoor informal learning environments. Common strategies—such as group inquiry [17], curiosity stimulation [13], and collaborative sense-making [31]—are effective in promoting cognitive engagement and parent-child interaction. In settings like museums and zoos, these practices rely on parents guiding attention, interpreting exhibits, and supporting exploration [30].

However, these strategies often assume full visual access—an assumption that fails in visually impaired families. Without real-time visual feedback, PVI struggle to observe and respond to their child's interests, disrupting joint attention [15]—a key to shared exploration. Moreover, common practices like pointing or demonstrating [20] become ineffective when visual cues cannot be shared, and children may not fully understand their parent's perceptual limits, leading to communication breakdowns. Additionally, most informal learning spaces are visually oriented in design—favoring sighted users through exhibit content [13], layout, and interaction mechanisms [10]. This vision-centric approach can exclude PVI, limiting their participation and marginalizing them within shared family experiences.

Thus, while current practices support most families, they fall short for those with visually impaired parents. These families face unique barriers—sensory asymmetry, limited access, and mismatched interaction modes. To ensure equity, new strategies and inclusive design approaches are needed to support meaningful, co-learning experiences for all.

3 Method

This study used semi-structured interviews and non-participant observations to investigate how PVI interact with their sighted children in outdoor informal learning, focusing on interaction dynamics, challenges, and strategies. Non-participant observation refers to a method where the researcher observes the subjects in their natural setting without direct involvement, allowing for an unobtrusive understanding of real-life behaviors and contexts [1].

3.1 Participants

Nine PVI (5 fathers and 4 mothers) were recruited via blind communities and snowball sampling. Their children were aged 6–10 ($M = 7.78$), and the parents were aged 34–39 ($M = 36.0$). Families differed in blindness status (7 dual-blind, 2 single-blind), demographics, and family structure. The study was approved by the Ethics Committee at the authors' institution.

3.2 Interview Methods

Each parent completed a one-hour interview covering demographics, outdoor experiences, informal learning views and attitudes toward assistive technologies. Two families participated in 3.5-hour non-participant observations. Researchers accompanied them unobtrusively and later conducted a brief focus group to reflect on shared

moments. All participants provided informed consent. Guardians authorized the direct observation of children, and all data were anonymized.

3.3 Data Analysis

We conducted a thematic analysis following Braun and Clarke's approach [4], combining interview transcripts and observational field notes. A codebook was iteratively refined through peer debriefing [21], and all data were re-coded accordingly.

4 Findings

This section presents our key findings, organized into two dimensions: objective and subjective. On the objective level, the lack of visual ability creates various challenges for PVI in outdoor informal education settings, primarily due to structural and perceptual constraints inherent to visual impairment. These challenges are often fundamental and resistant to change without external intervention. Objective difficulties also impact PVI on a subjective level, influencing how they parent, communicate, and share learning with their children.

4.1 Objective Level: Constraints of Structural and Perceptual Conditions

Based on the "Brain-Body-Environment" model [3], we map the challenges PVI face in outdoor informal learning across cognitive, physical and cultural dimensions which prove difficult to address without external support.

4.1.1 Individual Knowledge Reserve Limitations One major challenge for PVI lies in the cognitive gap between them and their sighted children. Due to different visions, whether from birth or acquired later, education, and acceptance of new technology, PVI perceive and understand the world differently from sighted individuals. For example, PVI since birth may struggle to explain visual concepts to children. P3: *"My child once asked what 'black' looked like. Her dad, who was born blind, couldn't answer. But I lost my vision at 20, so I said 'black is the color of the night sky.'"* This illustrates how sensory differences lead to mismatched understandings, making it hard for children to receive expected explanations.

Another key challenge is PVI's limited access to educational content, as much of today's educational media still lacks accessibility. Such inaccessibility leaves even PVI with full special education feeling helpless and directly hinders their ability to support children's learning and respond to their questions. P1: *"She loves watching documentaries, but some are in English. I don't understand English, and it's really hard for us to learn. So when she asks, I can only guess."*

4.1.2 Physical Environment Perception Barriers Perceptual limitations caused by visual impairment significantly affect the quality of parent-child interaction and restrict parents' involvement in education. In visually dense environments like museums, children often struggle to continuously describe what they see, leading to frequent breakdowns in communication. Many parents said they mainly accompany their children for the child's learning while gaining little meaningful experience themselves. Furthermore, difficulty establishing joint attention often results in missed learning moments. P9: *"There are many paintings in the museum that I am*

familiar with, but when my child asks about it, I can't answer because I can't see which one." P7: *"I can't enjoy the exhibits myself. If she's really into something, she'll stop for a long time and I just wait patiently. If it takes too long, I'll call out to check on her."*

Safety concerns further restrict where they can comfortably go. After briefly losing her child in a crowded area, P8 developed a fear of open, busy environments and now avoids them, preferring enclosed or familiar spaces. Children's freedom is limited, with independence allowed only under strict rules, such as returning promptly when called.

4.1.3 Social Identity and Prejudice Constraints Despite increasing public awareness, the stigmatized identity of PVI shapes their social interactions and self-perception. Non-hostile but frequent gazes and comments reinforce a sense of otherness. In one observation (F1), a family consistently attracted attention, making the child aware of their perceived difference and potentially disrupting parent-child interaction.

In addition to public perceptions, PVI often encounter overprotective attitudes from family or community members. Well-intentioned relatives may question their parenting abilities, inadvertently diminishing their role. F2: *"I said in our family chat that I'd pick up my child from school—no one replied. But when my husband said so, everyone agreed immediately."* Such implicit distrust undermines parental confidence and limits parent-child outdoor experiences.

4.2 Subjective Level: Influence of Parents' Educational Beliefs

Based on empirical observations and practical constraints, we identified three educational patterns among PVI based on their level of involvement: over-involved, hands-off, and limited support. Each participant was positioned in a diagram (Figure 2-a) to illustrate this distribution.

4.2.1 Over-involved Parents with *over-involved* tendencies in informal learning often conflate "support" with "control", prioritizing standardized outcomes over the learning process or their child's actual needs. In one observation (F1), the child hesitated to climb rocks out of fear, but his parents—unable to visually assess the situation—insisted he continue. P2: *"He's too timid... I don't want to miss the chance to teach him to toughen up."*

This often stems from parental anxiety about their limitations. P3: *"I can't see what he's doing, so I have to keep reminding him what to do, otherwise I always feel like something might go wrong."* Without visual feedback or a sense of active involvement, they may overcompensate with stricter control and authoritative commands. Driven by concern for the child's future, they may overemphasize discipline, unintentionally limiting autonomy and emotional expression.

4.2.2 Hands-off Parents with *hands-off* tendencies often consciously withdraw from educational roles, believing that their non-intervention will not hinder their child's development—and may even encourage greater independence. P2: *"I was never good at studying, so I don't expect anything specific from my child."* P4: *"With today's advanced technology, kids can find the answers they want themselves."*

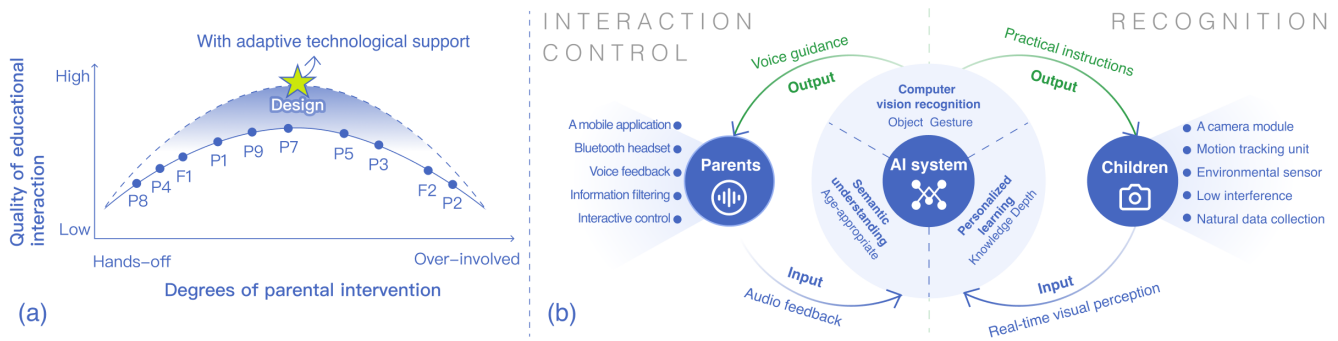


Figure 2: (a) A diagram mapping participants and highlighting our design goal; (b) Product Structure

This pattern stems from both functional and psychological constraints. Functionally, sensory limitations make it difficult for parents to access the full environmental context, leading to reduced engagement and little initiative in seizing educational opportunities. “Sometimes my child asks questions I can’t answer because I can’t see—like why a tree looks a certain way—so I usually let him figure it out himself.” Psychologically, feelings of self-doubt and guilt drive further retreat, with some parents internalizing the label of “non-traditional” or reframing withdrawal as fostering independence. P8: “I don’t worry too much about answers my child gets—no one is sure what’s right anyway. He’ll form his understanding.” Though appearing to respect autonomy, this approach reflects ultimately passive coping. Without adequate support, the child’s cognitive and emotional development may be compromised.

4.2.3 Limited Support Despite limited capabilities, many PVI actively provide timely, needs-based educational support. This reflects a *limited support* pattern, where parents assume selective responsibilities aligned with their abilities. For instance, P7 proactively chose recreational sites with educational value and encouraged his child to share observations and feelings to facilitate joint exploration.

This pattern is shaped by both a commitment to fostering independence and a clear awareness of personal limitations. P9: “When I’m out with my child if I miss someone doing good deeds, I lose the chance to give her real-time educational guidance.” Due to sensory constraints, PVI often face uncertainty about when and how to intervene. This may result in missed educational moments or low-quality involvement, revealing a clear bottleneck in their educational engagement.

4.3 Educational Model Design: Transitioning from Asymmetrical Dynamics to Symbiotic Education

In response to the asymmetrical parent-child dynamic, we propose the Symbiotic Education Model—a novel framework for PVI in outdoor informal learning. Centered on mutual understanding and support, it repositions PVI as learning facilitators empowered by assistive technologies to co-construct knowledge with their sighted children. Moving beyond hands-off or over-involved patterns, this model promotes shared developmental experiences (Figure 2-a).

Technology supports this relationship along two dimensions to disrupt the traditional one-way “teaching-learning” structure: enhancing parents’ contextual awareness to enable meaningful, responsive engagement and protecting children’s autonomy, preventing overdependence and intrusive involvement while supporting natural learning paths.

5 Design Framework

5.1 Design Implementation

To support Symbiotic Education Model, design should augment visual perspectives, enabling PVI to understand and participate in their child’s exploration, and fostering a co-constructive parent-child learning relationship. The design should follow these key principles:

The design should follow these key principles: (1) sense child behaviors—recognize the child’s natural intentions to help parents establish joint attention; (2) respect autonomy—offer lightweight, timely support without disrupting child-led exploration; (3) act as conversational scaffold [12]—support and encourage direct parent-child dialogue; (4) empower parents—provide flexible, adjustable feedback to support meaningful, non-controlling participation.

5.2 Product Forms

Unlike conventional all-in-one smart devices, our system uses a distributed sensing architecture (Figure 2-b), with the recognition module located on the child’s side. By capturing the child’s first-person perspective, the system treats their spontaneous gestures as entry points for exploration. These cues are processed and relayed to the parent, aligning their attention with the child and facilitating mutual understanding through responsive feedback.

5.3 Key Scenarios

The following four scenarios (Figure 1) illustrate how our design empowers PVI and their children in outdoor informal learning contexts.

Context-Aware Activation The device is activated through the child’s natural gestures and voice cues, automatically capturing and processing visual context. This enables real-time awareness while preserving the child’s freedom and allowing the parent to stay subtly engaged.

Parent-Controlled Feedback Once exploration is detected, semantic feedback is sent to the parent. Parents can adjust how and when they receive this input—modifying frequency, detail, or audio length via the app.

Educational Opportunity Guidance The system suggests additional educational content based on the object and the child's age or interests. This helps parents extend the conversation and guide deeper understanding.

Co-exploration The app provides a shared interface where parents and children can talk, reflect, and learn together—supporting real-time interaction and joint discovery.

6 Discussion and Conclusion

This study explores how perceptual asymmetries shape the dynamics of outdoor informal learning between PVI and their sighted children. We introduce Symbiotic Education Model that shifts assistive technology from individual compensation to relational empowerment—supporting mutual exploration, shared meaning-making, and more balanced parent-child collaboration.

Despite the insights offered, this study has several limitations. Due to the limited sample size and family variability, we cannot quantify the long-term effects of parenting strategies; our findings reflect structural tendencies and relational patterns rather than causal outcomes. Additionally, the complexity of children's outdoor behavior challenges real-time sensing, and the long-term relational impact of embedded technologies remains uncertain.

Future work should focus on prototyping, in situ testing, long-term observation of parenting dynamics, and developing privacy-aware systems. We also plan to adapt the framework to other forms of ability diversity.

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A Information of Participants

Parent ID	Parent Gender	Parent Age	Parent Work	Child Gender	Child Age	Both PVI or not	Only child or not
P1	Male	37	Traditional Chinese Massage	Female	6	T	T
P2	Male	35	Accessibility Developer	Male	7	T	T
P3	Female	39	Sensory Evaluator	Male	7	T	F
P4	Female	38	Postpartum Rehabilitation	Male	9	T	T
P5	Male	34	Salesman	Female	6	F	F
P6	Male	34	Traditional Chinese Massage	Male	10	F	F
P7	Male	34	Music Performer	Female	9	T	T
P8	Female	35	Traditional Chinese Massage	Female	7	T	T
P9	Female	38	Piano Tuner	Female	9	T	F
F1	/	40 / 37	Developer / Housewife	Male	9	T	T
F2	/	38 / 33	Repairman / Housewife	Male	6	F	T

Table 1: Information of Participants.P1–P9: individual interview participants; F1–F2: observation families